

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7.	(a)			<u>First finger</u> [in direction of magnetic] field / left to right / N to S (1) accept pointing finger <u>Second (middle) finger</u> [points in direction of] current / from B to A (1) <u>Thumb</u> [will give direction of] motion / up / force (1)	3			3		
	(b)			Gives a change of direction of current in the coil every half turn (1) So the force on the left is always up or the force on the left is in the same direction or the force on AB is upwards for half a turn and downwards for the other half turn (1) Award 1 mark for if not split ring it will only make half a turn	2			2		
	(c)	(i)		Selection of length of 6 (cm) (1) $F = 0.04 \times 0.7 \times 6 \times 10^{-2}$ (1 – conversion to m) $= 1.68 \times 10^{-3}$ [N] (1) Award 2 marks for an answer of 1.68×10^n or 3.36×10^{-3} [N] If length of 3 cm used answer of 8.4×10^{-4} [N] award 2 marks Award 1 mark for an answer of 8.4×10^n or 3.36×10^n [N]		3		3	2	
		(ii)		Reference to doubling (1) Reference to doubling again or four times the force (2)	1	1		2	1	
				Question 7 total	6	4	0	10	3	0